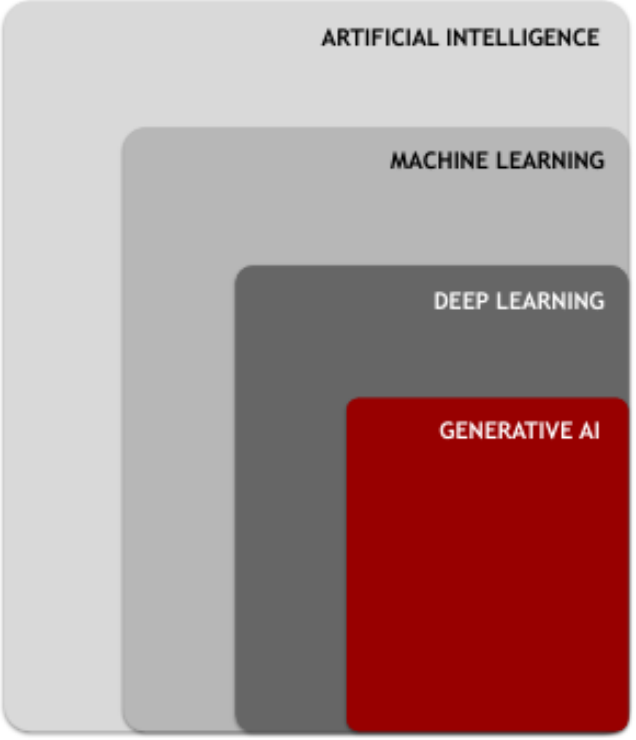
GenAI and LLM

Machine Learning is a type of AI, and Generative AI (GenAI) is a type of machine learning



1956 Class of problems we can solve when ***computers think/act like humans***

1997 ***Make predictions*** based on relationships inferred from historical data

2012 A ML Technique in which layer of ***neural networks are used to process data***

2020 ***Generate a wide variety of content*** using complex models trained on enormous datasets

Key ML Types

DISCRIMINATIVE MODEL

Used to classify or predict

GENERATIVE MODEL

Generates new data

Discriminative Model

Generative Language Model

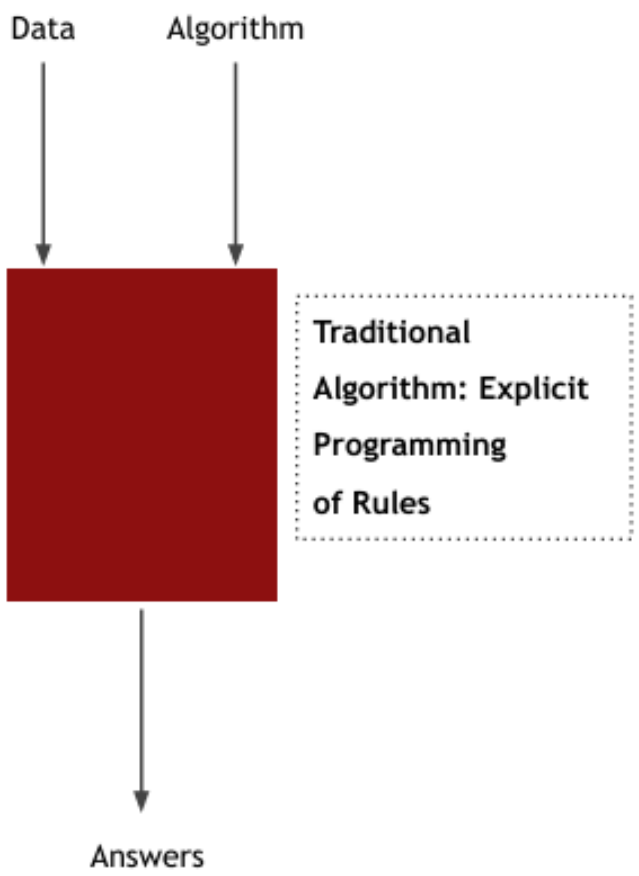
Read this huge pile of books on apples

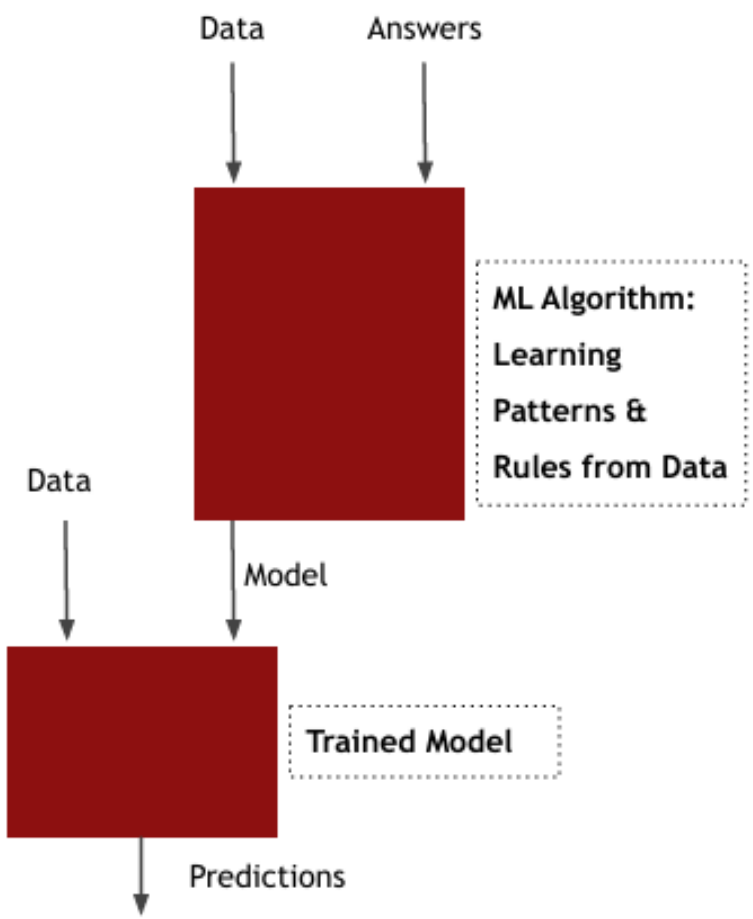
So, you have read about apple and its various types

Now tell me what is an apple?

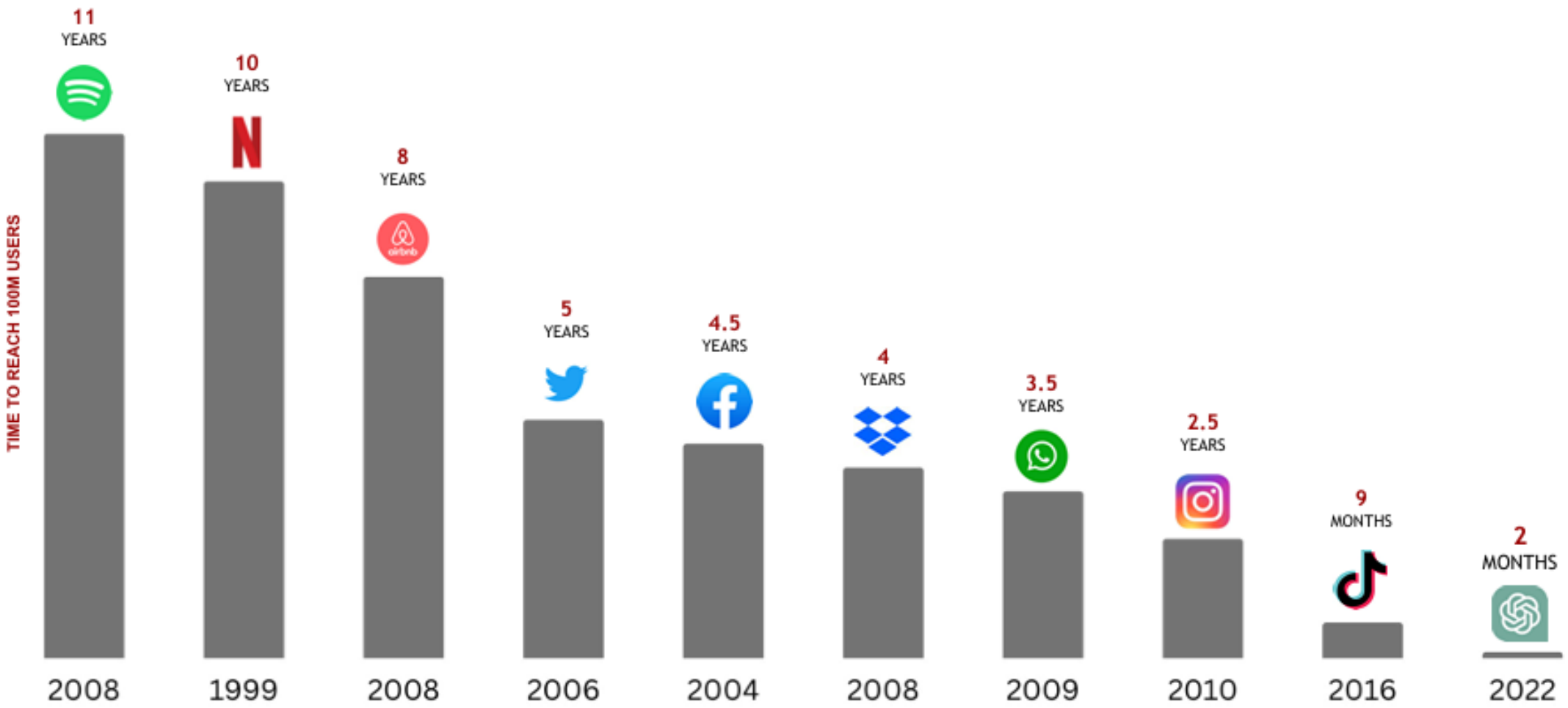
An apple is a ...

Traditional Systems vs Machine Learning based AI Systems



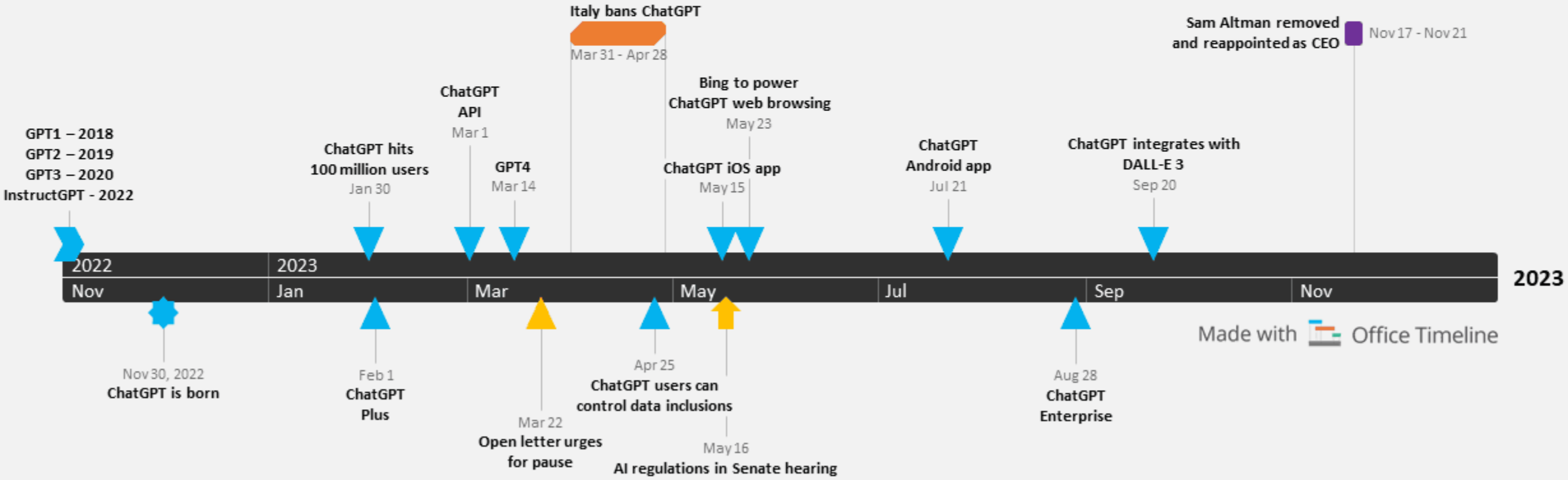


A New Wave!


TIME TO REACH 100M USERS
11 YEARS
10 YEARS
8 YEARS
5 YEARS
4.5 YEARS
4 YEARS
3.5 YEARS
2.5 YEARS
9 MONTHS
2 MONTHS
2008
1999
2008
2006
2004
2008
2009
2010
2016
2022

ChatGPT Timeline

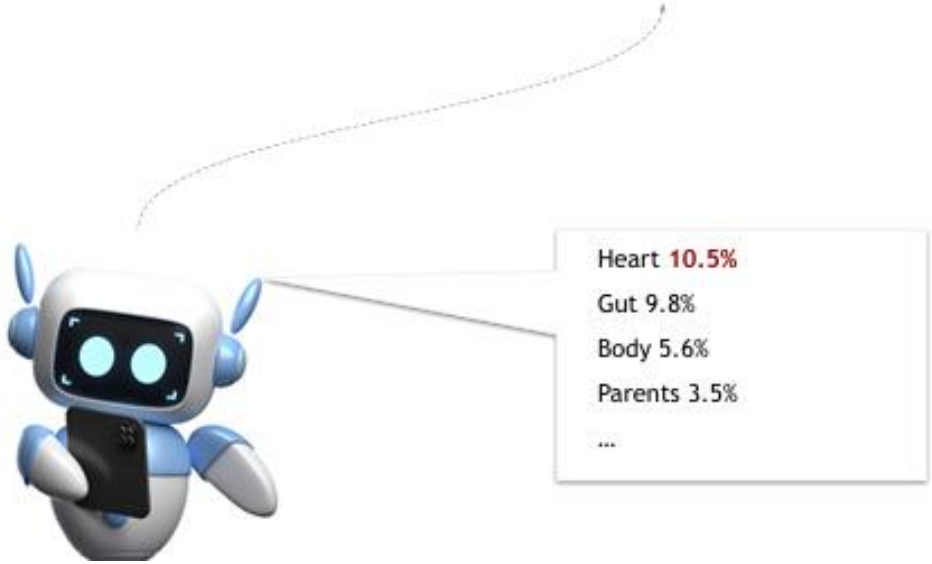
ChatGPT history and timeline



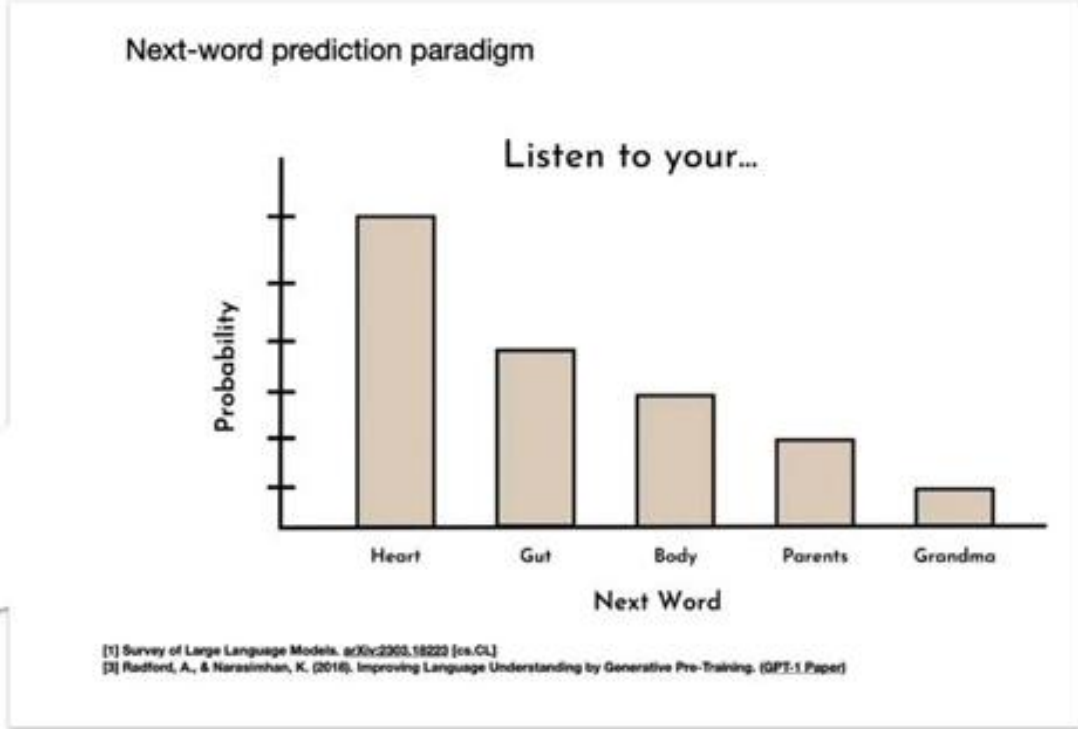
What is a Language Model?

A language model is like a ***smart machine that helps you finish sentences***. It's trained to predict and complete sentences, making it a super useful tool when you're typing or texting

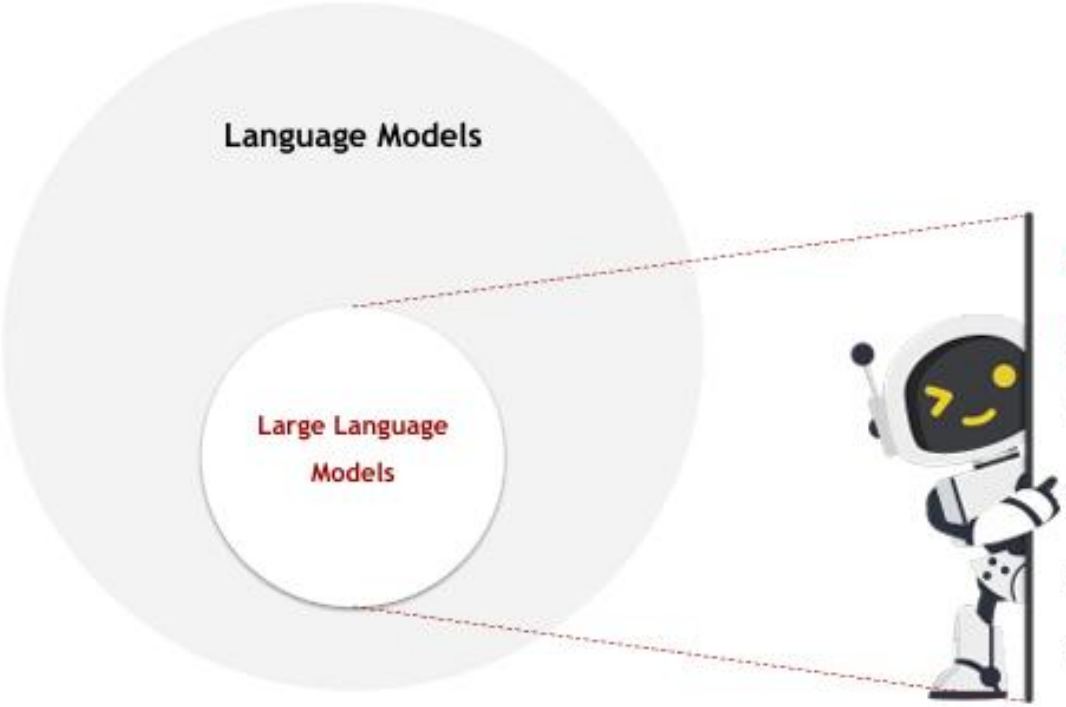
Listen to your _____



What is a Language Model?



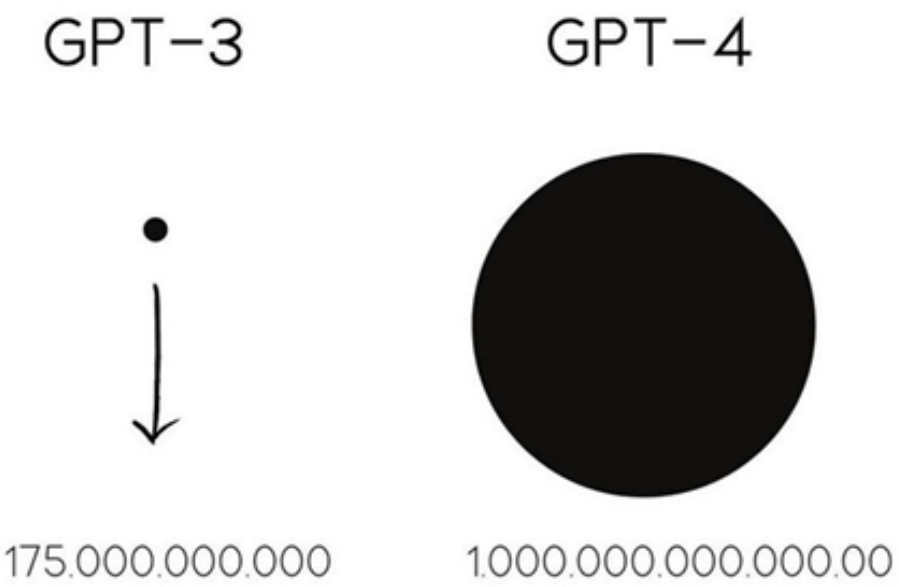
What are Large Language Models?



Models trained on large amounts of all type of publicly-available data that can perform a wide variety of tasks, including text/image/document summarization, generation, and categorization. These models can perform generative tasks like text/image/document generation and so there is some overlap between LLMs and Generative AI.

What is “**Large**” in Large Language Models?

- The term "**large**" refers to the size of the model in terms of its parameters and the volume of the training data
- Parameters are a part of the model that learns from historical training data
- They help in predicting the next word in a sequence
- One Trillion Parameters have been used in GPT 4



What is “**Language**” in Large Language Models?

- Human-understandable languages like English
- The models learn statistical patterns of the data
  - Example: Probability of "the" preceding the noun "cat" are very high
- These models can generate new text, mimicking Humans
- Applications:-

THANK YOU